Randomized Coverage Algorithms for Mobile Sensor Networks

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Abstract

In this work, without exploiting any location and distance information, distributed and randomized algorithms are devised for mitigating the un-coverage problem. These algorithms detect uncovered areas and redundant mobile sensor nodes, and direct them to move to cover the uncovered areas, only basing on the number of neighbors of each node by consuming a small amount of extra control packets. We also model mobile sensor networks as a random geometric graph and estimate the expected coverage ratio.

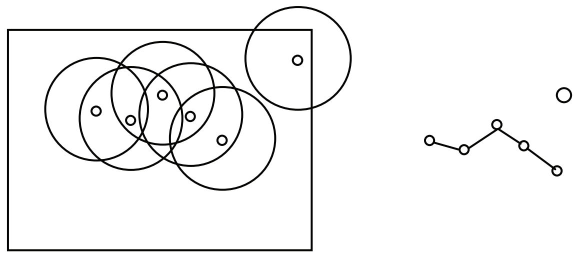
Keywords: Random geometric graphs, coverage problem, mobile sensor networks, randomized algorithms.

1. Introduction

One of many fundamental problems in wireless sensor networks is *the coverage problem* [2], which considers how to deploy or arrange sensor nodes in a desired area such that every point of that area is monitored by at least one sensor node for a period of time so that interested events or data can be thus collected and sent back to the back-end server for further processing. However, due to uneven deployment or failure of sensor nodes in sparse node density area, some locations might not be covered initially are never covered unless additional deployment (and extra costs) is activated.

To preserve the coverage of network, some researches suggest moving redundant mobile sensor nodes to uncovered areas for mitigating the un-coverage problem. Most of their approaches require location information of sensor nodes to calculate and determine redundant nodes, to detect uncovered areas, and to move those nodes to their pre-determined areas. Thus, their approaches assumed that each mobile sensor obtains its own location information through *global positioning system* (GPS), triangulation-based positioning protocols, or other location services. However, to acquire location information will increase the cost of hardware for deployment, extra computation and communication delay, and additional message overhead and power consumption. Some schemes adopt *received signal strength indicator* (RSSI) technology to calculate the distances between different nodes in the network. Moreover, most localization schemes and techniques become inaccurate or difficult to implement in an indoor area.

In this work, without exploiting any location and distance information, distributed randomized



(a) (b)

Figure 1. (a) A sensor network N=(6, r, A), where A is a rectangle. (b) Its associated random geometric graph $\Psi(X_6, r, A)$.

algorithms are devised for mitigating the un-coverage problem. The algorithms detect uncovered areas, redundant mobile sensor nodes, and direct them to move to cover the uncovered areas, only basing on the number of neighbors of each node and thus consuming a small amount of extra control packets. Moreover, we also estimate the expected coverage ratio of a given mobile sensor network by exploiting random geometric graph techniques.

A *geometric graph* $G=(V, r)$ consists of nodes placed in 2-dimension space R^2 and edge set $E=\{(i, j) \mid d(i, j) \le r$, where $i, j \in V$ and $d(i, j)$ denotes the Euclidian distance between node i and node $j\}$. Let $X_n=\{x_1, x_2, \ldots, x_n\}$ be a set of independently and uniformly distributed random points. We use $\Psi(X_n, r, A)$ to denote the *random geometric graph* (RGG) [8] of n nodes on X_n with radius r and placed in an area A. RGGs consider geometric graphs on random point configurations. Applications of RGGs include communications networks, classification, spatial statistics, epidemiology, astrophysics, and neural networks [8].

Here we use RGG to represent a mobile senor network. A mobile sensor network $N=(n, r, A)$ consist of n mobile devices with transmission radius r unit length that are independently and uniformly distributed at random in an area A. When each vertex in $\Psi(X_n, r, A)$ represents a mobile device, each edge connecting two vertices represents a possible communication link as they are within the transmission range of each other. In this work, we assume that the communication range of each sensor equals to its sensing range for simplicity. A random geometric graph and its representing network are shown in Figure 1. In the example, area A is a rectangle that is used to model the deployed area such as a meeting room. Area A, however, can be a circle, or any other shape, and even infinite space.

The rest of the paper is organized as follows. In Section 2, we briefly survey related results on RGGs. Distributed algorithms for the coverage problem is presented in Section 4, followed with its mathematic analysis in random geometric graphs. Section 5 presents how to modify the distributed algorithm into randomized algorithms for the coverage problem. Finally, Section 6 concludes the paper.

2. Related work in RGG

To the best of our knowledge, previous results on RGGs are all asymptotic and approximate except [12, 13]. We summary related results as follows.

A book written by Penrose [8] provides and explains the theory of random geometric graphs. Graph problems considered in the book include subgraph and component counts, vertex degrees, cliques and colorings, minimum degree, the largest component, partitioning problems, and connectivity and the number of components.

For n points uniformly randomly distributed on a unit cube in $d\geq 2$ dimensions, Penrose [9] showed that the resulting geometric random graph G is k-connected and G has minimum degree k at the same time when $n\to\infty$. In [3, 4], Díaz *et al.* discussed many layout problems including minimum linear arrangement, cutwidth, sum cut, vertex separation, edge bisection, and vertex bisection in random geometric graphs. In [5], Díaz *et al.* considered the clique or chromatic number of random geometric graphs and their connectivity.

Some results of RGGs can be applied to the connectivity problem of ad hoc networks. In [10], Santi and Blough discussed the connectivity problem of random geometric graphs $\Psi(X_n, r, A)$, where A is a d-dimensional region with the same length size. In [1], Bettstetter investigated two fundamental characteristics of wireless networks: its minimum node degree and its k-connectivity. In [6], Dousse *et al.* obtained analytical expressions of the probability of connectivity in the one dimension case. In [7], Gupta and Kumar have shown that if $r=\sqrt{\dfrac{\log n + c(n)}{\pi n}}$, then the resulting network is connected with high probability if and only if $c(n)\to\infty$. In [11], Xue and Kumar have shown that each node should be connected to $\Theta(\log n)$ nearest neighbors in order that the overall network is connected.

Recently, Yen and Yu have analyzed link probability, expected node degree, and expected coverage of MANETs [12]. Latter, Yen *et al.* have analyzed the k expected coverage problem in wireless sensor networks [13].

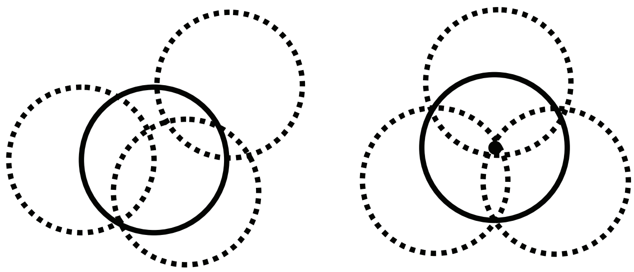
3. Distributed algorithms for detecting uncovered area and redundant mobile nodes

In this section, we first propose distributed algorithms for detecting uncovered area and redundant mobile nodes, followed with mathematical analysis.

Distributed coverage algorithms:

(1) The calling stage: Each node v decides whether it is on the boundary of an uncovered area by counting the number of its neighboring nodes $n(v)$. When $n(v)\leq\alpha$ then v has high probability to be on the boundary of an uncovered area. Node v issues a control packet over the network for calling other redundant mobile nodes to cover the uncovered area.

(2) The called stage: When a mobile node receives a control packet for calling redundant mobile nodes, the node v decides whether it is redundant by counting the number of its neighboring nodes $n(v)$. When $n(v)\geq\beta$ then v has high probability to be redundant and can move (after registering to the initiated node) to other places for covering uncovered areas.



(a) (b)

Figure 2. A circle (in solid line) can be covered by at least three neighboring circles if arranged properly.

The above coverage algorithms are simple and straightforward. Since it does not exploit any location and distance information such GPS, these algorithms are easy to be implemented distributedly or applied in indoor area.

The most difficult parts in designing the algorithms are how to select proper threshold values for α and β because different assignments of α and β will dramatically affects the performance of the proposed algorithm. Evidently, we can expect that $\beta > \alpha$ due to the following facts:

(1) When a node is located in a boundary of an uncovered area, it does not have much neighboring nodes.
(2) On the other hand, when a node *v* is redundant, its neighboring nodes fully cover the area where it located even without *v*. That indicates that when the number of its neighboring nodes is increased, node *v* is more likely to be redundant.

Since in our proposed algorithms each node *v* detects uncovered areas and redundant mobile sensor nodes only depending on the number of neighboring nodes, such limited information makes accurate detections impossible. For example, there maybe exist some nodes covering *v*, but they cannot directly communicate to node *v* (Figure 2(a)). As a result, these nodes cannot be detected by node *v* and we will lose their information in the proposed algorithms; in other words, these nodes are invisible to node *v*. Therefore, hereafter the neighboring nodes of *v* here indicate that these nodes can directly communicate to *v*.

The following theorem is quite straightforward, but rather fundamental.

Theorem 1: A node *v* is on the boundary of an uncovered area if and only if the circumference of the representing circle is not fully covered by the neighboring nodes of *v*.

Suppose that there exist no two circles with the same center. As shown in Figure 2 (b), three neighboring dash-line circles (if arranged properly) are enough to cover the solid-line circle, but two is always not enough. Consequently, we have the following lemma.

Lemma 2: we have $\alpha \geq 2$ and $\beta \geq 3$.

If we can obtain the probability $Q(n)$ of *n* random neighboring nodes (of *v*) cover its circumference, the values of α and β then can be selected or determined accordingly and easily. The following theorem will be useful for deriving the desired result.

Theorem 3 [14]: The probability $P(a, n)$ that n random arcs of angular size a cover the circumference of a circle completely (for a circle with unit circumference) is

$$P(a,n)=\sum_{k=0}^{\left\lfloor \frac{1}{a} \right\rfloor}(-1)^k\binom{n}{k}(1-ka)^{n-1}.$$

First, we derive a low bound for $Q(n)$ by using Theorem 3.

Theorem 4: We have $Q(n)\geq P(2\pi/3, n)$.

Proof: Since every neighboring node u of a node v must communicate to v directly, we have the center of the circle representing u (the dash circle in Figure 3) must be in the circle representing v (the solid circle in Figure 3). The part of circle v covered and contained by circle u is the arc of Angle $\angle aO_vb$, which must be greater than or equal to $2\pi/3$. Consequently, we have $Q(n)\geq P(2\pi/3, n)$.

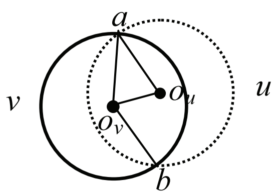


Figure 3. The part of circle *v* covered and contained by circle *u* is the arc of Angle $\angle aO_vb$, which must be greater than or equal to $2\pi/3$.

The main theorem is described as follows.

Theorem 5: We have $Q(n)=P(2\cos^{-1}(I_e/2r), n)$, where I_e is the expected distance of two centers of two neighboring nodes and r is the radius of a circle.

Proof: Since the length of interval $\overline{a,o_v}=\overline{a,o_u}=r$, if we can obtain the length I of interval of $\overline{o_v,o_u}$, then $\angle aO_vb=2\sin^{-1}(I/2r)$. Suppose that the expected distance of two centers of two neighboring nodes u and v is I_e, then the expected angle $\angle aO_vb=2\cos^{-1}(I_e/2r)$, then $Q(n)=P(2\cos^{-1}(I_e/2r), n)$.

When one of two equal-sized circles in the place contains the center of the other, we call them two *properly intersecting circles*. The remaining effort is to derive the expected distance of two centers of two neighboring nodes as shown in the next theorem.

Theorem 6: The expected distance of the centers of arbitrary two properly intersecting circles with the same radius r is $2r/3$ in a $\Psi(X_n, r, A)$.

Proof: Suppose that two circles of radius r centered at O and O' are properly intersecting. Let A and B be two distinct intersecting points of these two circles (See Figure 4)

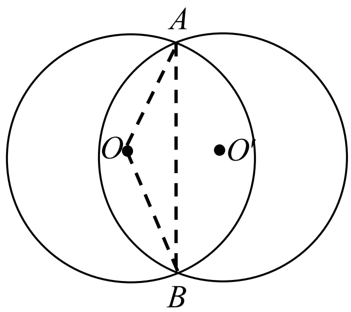


Figure 4. Two properly intersecting circles.

Let X be the distance between O and O'. Since the given two circles are properly intersecting, we have $X\leq r$ and $\Pr(X\leq r)=1$. The probability that X is less than x (i.e. $\Pr(X\leq x)$ where $x<=r$) is proportional to the area of the circle of radius x centered at O. Let $F(x)$ be the probability distribution function of X. Then we have

$$F(x) = \Pr\left[X \le x\right] = \frac{\pi x^2}{\pi r^2} = \frac{x^2}{r^2}.$$

Since $x= d(O, O')=2r\cos(\theta/2)$, the probability distribution function of θ is

$$\begin{aligned} G(y) &= \Pr\left[\frac{2\pi}{3} \le \theta \le y\right] \\ &= \Pr\left[2r\cos\frac{y}{2} \le x \le r\right] \\ &= F(r) - F\left(2r\cos\frac{y}{2}\right) = \text{-}2\cos\ y \text{ - } 1. \end{aligned}$$

Thus the probability density function of θ is $g(y)=G'(y)=2\sin y$.

Finally, the expected center distance of these two properly intersecting circles is

$$\begin{aligned} &\int_{\frac{2\pi}{3}}^{\pi} 2r\cos(\frac{\theta}{2})2\sin\theta\, d\theta \\ =& \int_{\frac{2\pi}{3}}^{\pi} 4r\cos(\frac{\theta}{2})\sin\theta\, d\theta \\ =& \int_{\frac{2\pi}{3}}^{\pi} 4r\cos(\frac{\theta}{2})2\cos(\frac{\theta}{2})\sin(\frac{\theta}{2})d\theta \\ =& \int_{\frac{2\pi}{3}}^{\pi} 8r\cos^2(\frac{\theta}{2})\sin(\frac{\theta}{2})d\theta \\ =& \int_{\frac{2\pi}{3}}^{\pi} 8r\cos^2(\frac{\theta}{2})(\text{-}2)(\text{-}\frac{1}{2})\sin(\frac{\theta}{2})d\theta \\ =& \int_{\frac{2\pi}{3}}^{\pi} (\text{-}16)r\cos^2(\frac{\theta}{2})d\cos(\frac{\theta}{2}) \\ =& (\text{-}16r)\left[\frac{1}{3}\cos^3(\frac{\theta}{2})\right]_{\frac{2\pi}{3}}^{\pi} \\ =& (\text{-}16r)(\frac{1}{3})((0)^3 \text{ - } (\frac{1}{2})^3) \\ =& \frac{2}{3}r \end{aligned}$$

By combing Theorem 5 and 6, we obtain the final theorem.

Theorem 7: We have $Q(n) \approx P(\frac{141}{180}\pi, n)$.

Proof: Since the expected distance of two centers of two neighboring nodes is $(2/3)r$ by Theorem 6, we have the expected angle $=2\cos^{-1}(I_e/2r)=$ $2\cos^{-1}((2/3)r/2r)=2\cos^{-1}(1/3) \approx (141/180)\pi$ by Theorem 5 and finally we have $Q(n) \approx P(\frac{141}{180}\pi, n)$.

4. Randomized coverage algorithms for detecting uncovered area and redundant mobile nodes

We modify the above distributed coverage algorithms into randomized coverage algorithms in this section.

Randomized coverage algorithms:

(1) The calling stage: Each node v decides whether it is on the boundary of an uncovered area by first counting the number of its neighboring nodes $n(v)$. We roll a dice such that the probability of deciding that v is on the boundary of an uncovered area is $1\text{-}P(\frac{141}{180}\pi, n(v))$. If node v is decided to be on the boundary of an uncovered area, node v starts to issues a control packet over the network for calling other redundant mobile nodes to cover the uncovered area.

(2) The called stage: When a mobile node u receives a control packet for calling redundant mobile nodes, the node decides whether it is redundant by counting the number of its neighboring nodes $n(u)$ first. Then roll a dice such that the probability of deciding that u is redundant is $P(\frac{141}{180}\pi, n(u))$. If node u is decided to be redundant, then it can move (after registering to the

initiated node) to other places for covering uncovered areas.

5. Conclusions

In this work, we exploit random geometric graph technique to design and analyze the proposed distributed and randomized algorithms for the coverage problem in wireless sensor networks. These algorithms make decision only depending on the number of neighboring sensor nodes, thus reducing the required hardware cost, system overhead, and communication delay. In the future, we try to improve and analyze the performance of these randomized algorithms if additional information such as angle information is supported.

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